ModelIt!

Teacher's Guide

Can We 3D-Print a Human Organ?

Modeling Bioprinting, Tissue Engineering, and Transplant Viability

Quick Reference

Grade Level	Grade 11L3
NGSS Standard	HS-LS1-2, HS-ETS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-2, HS-ETS1-2

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models of bioprinting and tissue engineering to predict how changes in print parameters, scaffold design, and vascularization affect the viability and function of bioprinted tissue constructs.
Disciplinary Core Idea	**LS1.A Structure and Function** — Students model the hierarchical organization of cells into tissues and organs, understanding how the structure of bioprinted constructs at the cellular and vascular level determines whether the tissue can perform its biological function.
Crosscutting Concept	**Structure and Function** — Students analyze how the three-dimensional architecture of bioprinted tissue — cell arrangement, scaffold geometry, blood vessel network — directly determines whether the construct can sustain cell life and replicate native tissue function.

CAST Testing Alignment

MULTIPLE CHOICE:

A bioprinted cardiac patch is 5 millimeters thick and contains printed blood vessel channels spaced 400 micrometers apart. Given the 200-micrometer nutrient diffusion limit, will all cells in the construct receive adequate oxygen and nutrients? Explain your reasoning using the relationship between diffusion distance and channel spacing.

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why bioprinting a functional skin graft is achievable with current technology while bioprinting a functional kidney remains decades away. Reference at least three model variables — including vascularization depth, cell type complexity, and functional maturation requirements — in your response.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Over 100,000 people in the United States are on the organ transplant waiting list right now, and 17 of them die every day because a matching donor organ is not available in time. This shortage exists because organ donation depends on tragic circumstances — healthy organs from recently deceased individuals whose families consent to donation — and the demand far exceeds the supply. Bioprinting offers a revolutionary alternative: manufacturing replacement organs from the patient's own cells, eliminating both the donor shortage and the risk of immune rejection. The technology has made remarkable progress in simple tissues, but the leap to complex organs faces fundamental biological barriers that no engineering solution has yet overcome.

How Bioprinting Works

Bioprinters are specialized 3D printers that deposit living cells rather than plastic. The most common approach — extrusion bioprinting — pushes bioink (cells suspended in hydrogel) through a fine nozzle to create lines of material that are stacked layer by layer. The nozzle diameter (typically 100-500 micrometers) determines print resolution, while the hydrogel's properties must be carefully balanced: viscous enough to hold shape after deposition but soft enough that shear forces in the nozzle do not kill cells. Other approaches include inkjet bioprinting (which fires tiny droplets of cell-laden fluid), laser-assisted bioprinting (which uses a laser to propel cells from a donor ribbon onto a substrate), and stereolithographic bioprinting (which uses light to crosslink photosensitive hydrogels containing cells). Each method offers different trade-offs between resolution, speed, cell viability, and scalability. After printing, the construct is placed in a bioreactor that provides nutrients, oxygen, temperature control, and sometimes mechanical stimulation during weeks of tissue maturation.

The Vascularization Challenge

The human body's blood vessel network is a fractal branching system spanning from the 2.5-centimeter-diameter aorta down to capillaries just 5-10 micrometers in diameter — smaller than a red blood cell. This network ensures that no cell is more than 200 micrometers from a capillary, because oxygen and nutrients can only diffuse that far through tissue before being consumed. In bioprinted constructs, cells more than 200 micrometers from a nutrient source begin dying within hours. For thin tissues like skin (1-3 millimeters thick), surface diffusion can sustain all cells during maturation. For thick tissues and organs, an internal blood vessel network is absolutely required. Current bioprinting can create channels for large and medium vessels (down to about 200-500 micrometers diameter), but the capillary-level branching that connects these channels to every cell requires self-assembly by the cells themselves — a process that takes weeks and is not reliably controlled. This vascularization gap is the single largest barrier between current tissue engineering and functional organ bioprinting.

Cell Sourcing and Immune Compatibility

Bioprinted tissue requires billions of living cells. For a heart, that number is approximately 2-3 billion cardiomyocytes plus billions more endothelial cells, fibroblasts, and smooth muscle cells. These cells can come from the patient (autologous), from donors (allogeneic), or from pluripotent stem cell lines (iPSCs). Autologous cells — derived from the patient's own tissue and reprogrammed into needed cell types using induced pluripotent stem cell technology — eliminate immune rejection but require 4-8 weeks of cell culture to produce sufficient quantities. Donor cells and stem cell lines are more quickly available but trigger immune responses that may require lifelong immunosuppression. The iPSC approach is currently the most promising: a small skin biopsy from the patient is reprogrammed into stem cells that can differentiate into any cell type needed for the organ. However, iPSC reprogramming is expensive, takes weeks, and carries a small risk of genetic mutations during the reprogramming process.

Current Achievements and Future Trajectory

As of 2025, bioprinting has achieved clinical or near-clinical success in several tissue types. Bioprinted skin grafts are in clinical trials for burn treatment. Bioprinted cartilage implants have been successfully implanted in human ears and noses. Researchers at the Wake Forest Institute for Regenerative Medicine have bioprinted bladder scaffolds that were successfully implanted in patients. Bioprinted blood vessel segments have been tested in animal models. Organovo has bioprinted liver tissue patches

for drug toxicity testing. However, no complete solid organ has been bioprinted and transplanted. The progression from simple tissues to complex organs is not linear — it requires simultaneous breakthroughs in vascularization at capillary scale, multi-cell-type coordination, functional maturation of specialized structures (like kidney nephrons or liver lobules), and quality control for constructs containing billions of cells. Most researchers estimate that bioprinted tissue patches for organ repair will become clinically routine within 10-15 years, while complete bioprinted organs are 20-30 or more years away.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify seven key components of a bioprinting system: print resolution, cell viability rate, vascularization depth, scaffold degradation rate, nutrient diffusion limit, immune compatibility, and functional maturation.

E **Establish Relationships**
Students map relationships showing that nutrient diffusion limit and vascularization depth are the primary constraints determining maximum construct thickness, while print resolution and cell viability determine structural fidelity, and scaffold degradation must be matched to functional maturation timing.

V **Visualize & Model**
Students build a computational model showing the seven-component bioprinting system with cell survival and tissue function as dependent outcomes.

E **Evaluate Outcomes**
Students run skin patch, cardiac patch, and full kidney scenarios to discover how complexity scaling reveals the vascularization barrier as the central unsolved challenge.

R **Revise & Extend**
Students add multi-material printing, innervation, or mechanical conditioning to model more realistic tissue engineering complexity and functional requirements.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Seventeen people die every day in the United States waiting for an organ transplant. What if we could print new organs from the patient's own cells — no donor needed, no rejection, no waiting list? That is the promise of bioprinting. The reality is more complicated — and more fascinating — than it sounds."*

Watch For

Open with the transplant shortage statistics. Ask: who knows someone who has been affected by organ failure or the transplant waiting list? Establish the human stakes before the science.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the intersection of 3D printing technology and living biology. You will discover why printing a skin graft is hard, printing a heart patch is harder, and printing a complete kidney is a challenge that may take decades to solve."*

Watch For

Have students read objectives. Pre-teach 'bioink' and 'vascularization.' Demonstrate the 200-micrometer diffusion limit with a visual: hold up a sheet of paper (100 micrometers thick) and explain that cells can only survive within two paper-thicknesses of a blood vessel.

Slide 3: Big Question

3 min

Say: *"We CAN print living tissue — several labs have done it. We can even implant bioprinted tissue into patients. The question is whether we can scale from millimeter-thin patches to centimeter-thick organs with billions of cells, miles of blood vessels, and dozens of specialized cell types. The biology says: not yet."*

Watch For

Quick-write: Students estimate how many cells are in a human heart and how many blood vessels supply them. Reveal the real numbers (2-3 billion cells, 100,000+ miles of capillaries) and discuss the engineering scale.

Slide 4: LEVER Framework

2 min

Say: *"We are modeling a system where engineering precision meets biological complexity, and where the laws of physics — specifically diffusion — set hard limits on what is possible."*

Watch For

Preview LEVER steps. Emphasize that this model will reveal why the jump from tissue patches to complete organs is not gradual but requires qualitative breakthroughs.

Slide 5: Activity 1: Components

8 min

Say: *"Seven components. Two are physical constraints you cannot change: print resolution and the 200-micrometer nutrient diffusion limit. The other five are biological variables that determine whether your printed tissue lives or dies."*

Watch For

Guide through components. Use a microscope image to show the scale of 200 micrometers. Discuss why print resolution and nutrient diffusion are external constraints while the biological responses are internal.

Slide 6: Activity 2: Connections

8 min

Say: *"Everything flows through vascularization. No blood vessels, no oxygen. No oxygen, dead cells. Dead cells, failed organ. The entire challenge of organ bioprinting reduces to one question: can you print a blood vessel network at capillary scale?"*

Watch For

Help students map the critical vascularization pathway. Calculate how many capillaries a kidney needs to serve its million nephrons. Highlight the scaffold degradation timing challenge.

Slide 7: Activity 3: Simulation

10 min

Say: *"Three tissues, three levels of complexity, three very different outcomes. Watch what happens when you ask the same system to do progressively harder things."*

Watch For

Students predict viability and function before each scenario. The skin scenario succeeds, the cardiac patch partially succeeds, and the kidney scenario reveals why full organs remain beyond reach. Discuss the qualitative difference between incremental and breakthrough challenges.

Slide 8: Discoveries

5 min

Say: *"Bioprinting is not just a printing problem — it is a biology problem. The printer deposits cells, but the cells must organize themselves, build their own structures, and become a functioning tissue. We can guide this process but not fully control it."*

Watch For

Lead discussion on the difference between engineering (precise control) and biology (guided self-organization). Why does tissue maturation take weeks? Why is the outcome variable between batches? Connect to the broader theme of engineering living systems.

Slide 9: STEM Challenge

12 min

Say: *"A hospital needs your tissue engineering design for a real clinical problem. Choose your application, design the construct, and prove it will work."*

Watch For

Groups choose between skin, cartilage, or cardiac applications and design complete bioprinting protocols. Must specify cell source, bioink, scaffold, vascularization strategy, maturation protocol, and quality tests. Present with model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Print Resolution, Nutrient Diffusion Limit
- **INTERNAL:** Cell Viability Rate, Vascularization Depth, Scaffold Degradation Rate, Immune Compatibility, Functional Maturation

Relationships

- **Print Resolution to Cell Viability Rate:** POSITIVE (+) up to optimum, then NEGATIVE (-) — Higher print resolution initially improves viability by creating more precise cell arrangements and better nutrient channel geometry. However, achieving very high resolution requires smaller nozzle diameters that increase shear stress on cells, eventually reducing viability. The relationship peaks at the optimal balance between structural precision and mechanical cell damage.
- **Vascularization Depth to Functional Maturation:** POSITIVE (+) — Greater vascularization depth means more cells throughout the construct receive adequate oxygen and nutrients, enabling them to survive, proliferate, and develop the specialized functions of mature tissue. Without sufficient vascularization, cells in the interior die before they can mature.
- **Nutrient Diffusion Limit to Vascularization Depth:** NEGATIVE (-) — The nutrient diffusion limit constrains vascularization depth because it sets the maximum distance between blood vessels — if channels are spaced farther apart than twice the diffusion limit, cells between them die. This constraint forces denser vascular networks as construct thickness increases.
- **Scaffold Degradation Rate to Functional Maturation:** NONLINEAR (optimal range) — If scaffold degradation is too fast, the construct loses mechanical support before cells produce sufficient extracellular matrix, causing collapse. If too slow, the persistent scaffold physically prevents cell migration, matrix deposition, and tissue remodeling. Only when degradation rate matches maturation rate does the tissue successfully transition from scaffold-dependent to self-supporting.

Key Discoveries

- The 200-micrometer nutrient diffusion limit is the fundamental constraint that makes organ bioprinting qualitatively different from printing simple tissues — every thick construct must include a functional vascular network, and printing functional blood vessels at capillary scale remains the greatest unsolved challenge
- Cell viability decreases with print time because cells outside physiological conditions accumulate damage — this creates a race between printing speed and cell death that becomes more severe as constructs become larger and more complex
- Scaffold degradation timing must be precisely matched to tissue maturation rate — a mismatch of even a few days can cause either structural collapse or prevented tissue remodeling, and this timing varies unpredictably between patients and cell batches
- The gap between current capability (thin tissue patches, cartilage, simple blood vessels) and the goal (complete transplantable organs) is not incremental — it requires solving vascularization, multi-cell-type coordination, and functional maturation simultaneously at a complexity level thousands of times greater than anything achieved to date

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Bioprinting organs works just like 3D printing plastic objects"	3D printing plastic is purely an engineering problem: deposit material, let it harden, done. Bioprinting deposits living cells that must survive the printing process, receive continuous nutrients and oxygen, organize themselves into functional tissue, and mature over weeks in bioreactors. The printed construct immediately after printing is like a building frame with workers inside — the structure exists but the function has not yet developed. Cell death begins within hours if nutrient supply is inadequate. The printed architecture may not match the final tissue architecture because cells migrate, proliferate, and remodel their environment. The model shows that printing is only about 10% of the challenge — the other 90% is keeping cells alive and guiding them to functional maturity.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"We will be able to print replacement organs within five years"	This misconception confuses the pace of progress in simple tissues with the timeline for complex organs. Bioprinted skin grafts and cartilage implants may reach routine clinical use within five years because they are thin, involve few cell types, and have relatively simple functional requirements. Complete solid organs require solving vascularization at capillary scale, coordinating dozens of cell types in precise three-dimensional arrangements, achieving functional maturation of complex structures like nephrons or cardiac conducting fibers, and producing billions of patient-specific cells. These are not incremental improvements but fundamental breakthroughs. The model quantitatively demonstrates the orders-of-magnitude gap between current tissue capability and organ-level complexity.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Using the patient's own cells means there is no risk of immune rejection"	While autologous cells (the patient's own cells) dramatically reduce rejection risk compared to donor cells, the bioprinting process introduces non-self materials — hydrogel scaffolds, crosslinking agents, residual growth factors — that can trigger immune responses. Additionally, the cell reprogramming process (iPSC generation) can introduce mutations that make cells partially foreign to the immune system. Scaffold degradation products may cause local inflammation. And if the bioprinted construct contains any contaminating bacteria or endotoxins from the manufacturing process, the immune response can be severe. The model shows that immune compatibility is a spectrum, not a binary, and that using autologous cells reduces but does not eliminate immune-related challenges.